



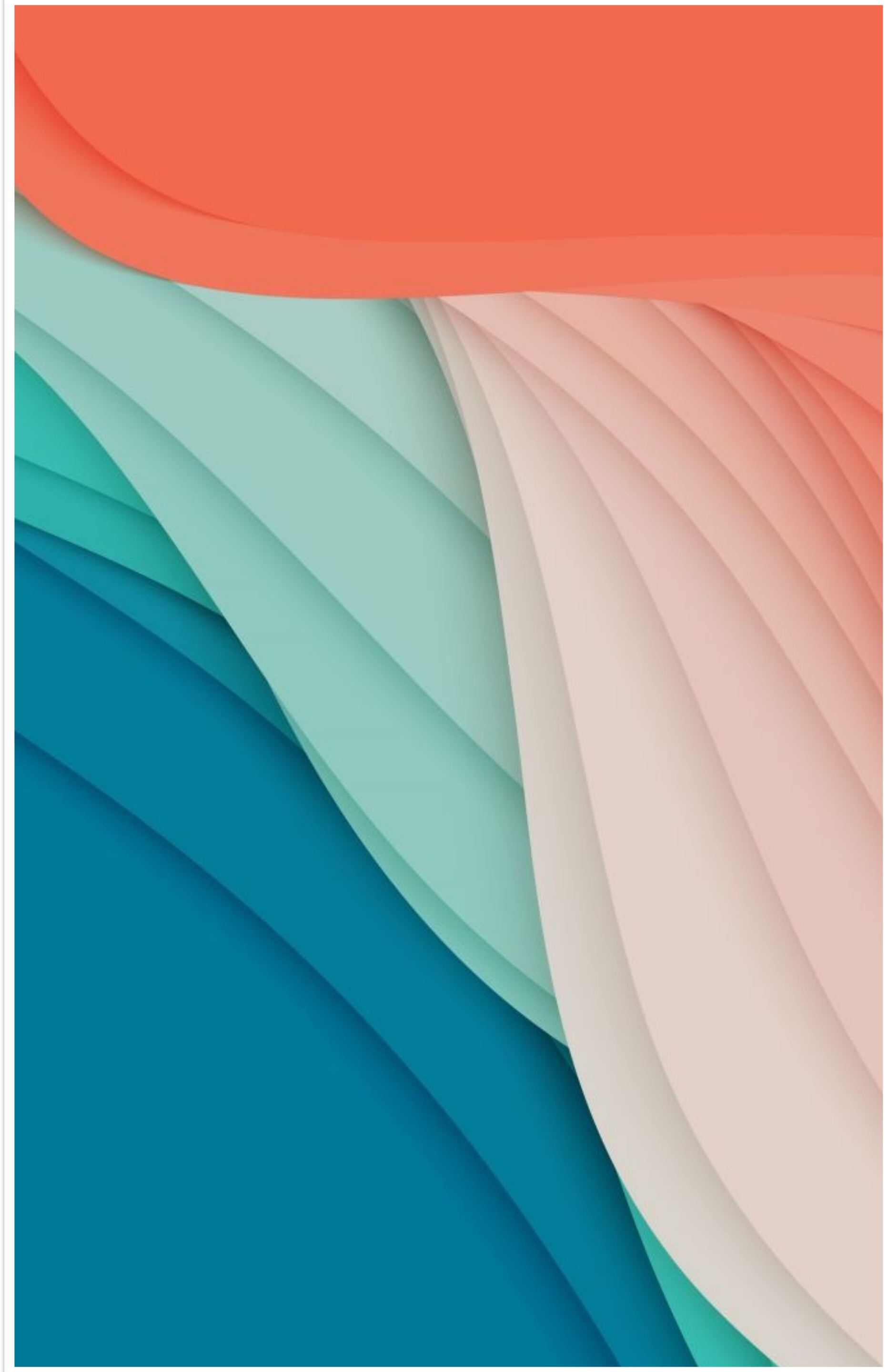
Git - Part I

Introduction & Basics

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Année universitaire 2026/2027



Contenu du cours

1. Git et github

- a. Initiation à git : commandes de base.
- b. Utilisation de Visual Studio Code.
- c. Branches dans git.
- d. Collaboration dans git (GitHub).

2. Docker

- a. Bases de docker.
- b. Utilisation de docker hub.
- c. Automatisation avec docker-compose.



Projet



Projet final



Projet final 2



Projet final 3



Projet final final



Projet final finaal



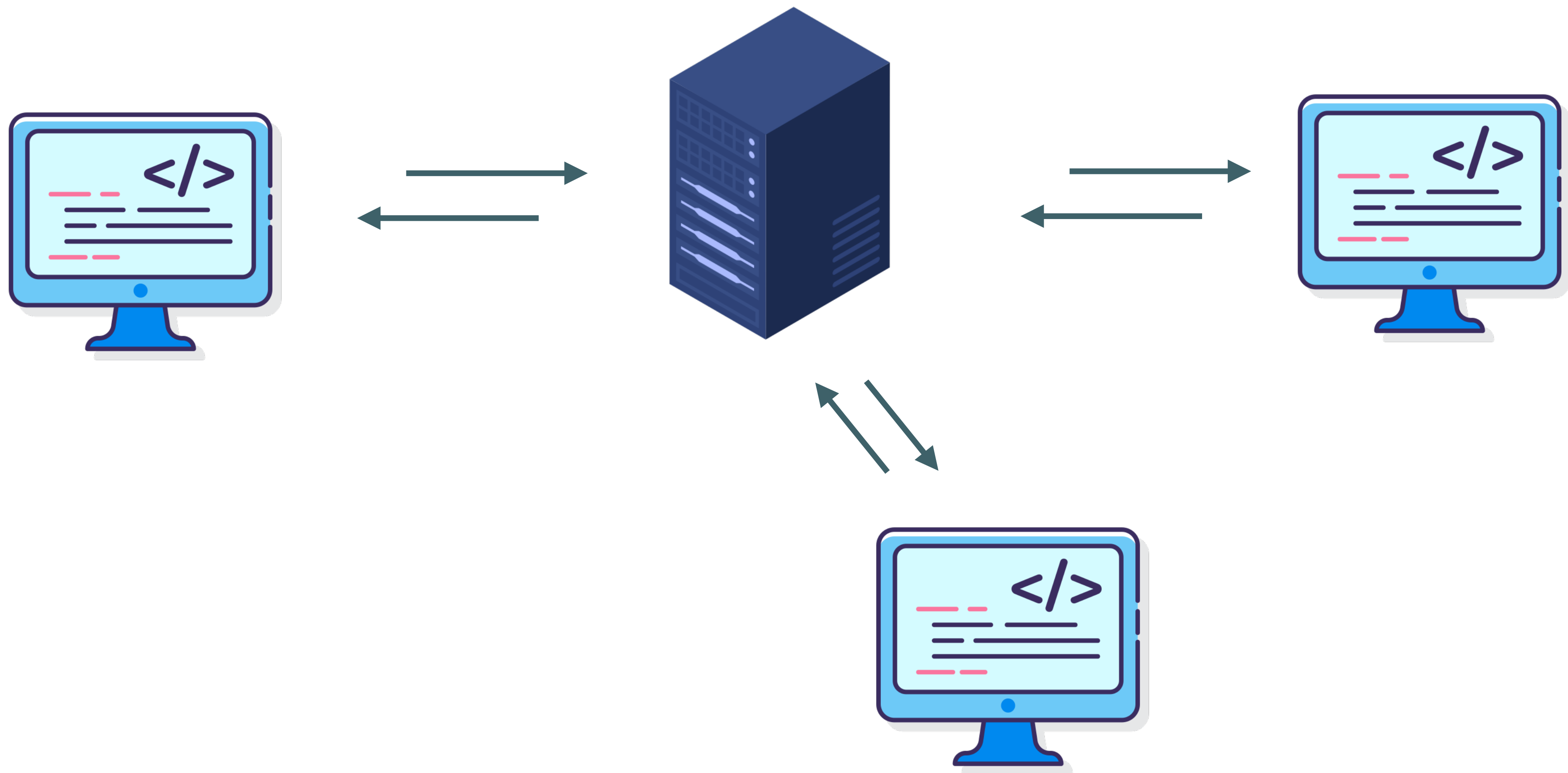
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What is Git ?

- ◆ Git is a **distributed** version control system (VCS).
- ◆ Track changes made on files and revert back to them (like a time traveling machine).
- ◆ Developed and released in 2005 by Linus Torvalds, the creator of Linux.
- ◆ It is a free and open source.
- ◆ According to a [Stack Overflow developer survey](#) over 87% of developers use Git.

What is a distributed VCS

No single point of failure

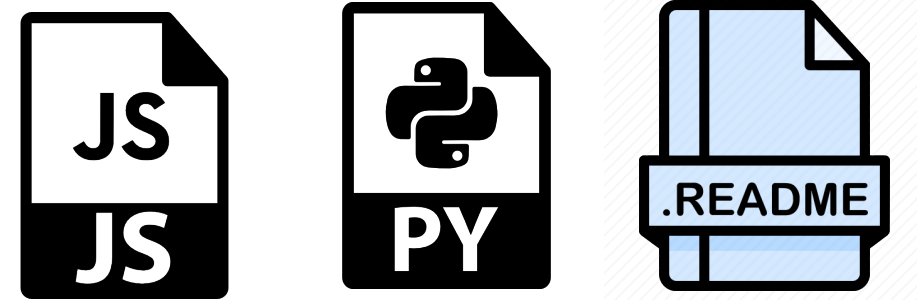


Git hosting service

- ◆ Web hosting service that hosts git projects on the cloud.
- ◆ The most used ones are : Github, bitbucket and gitlab.
- ◆ **Github** is the most popular one, it was bought by Microsoft in 2018.

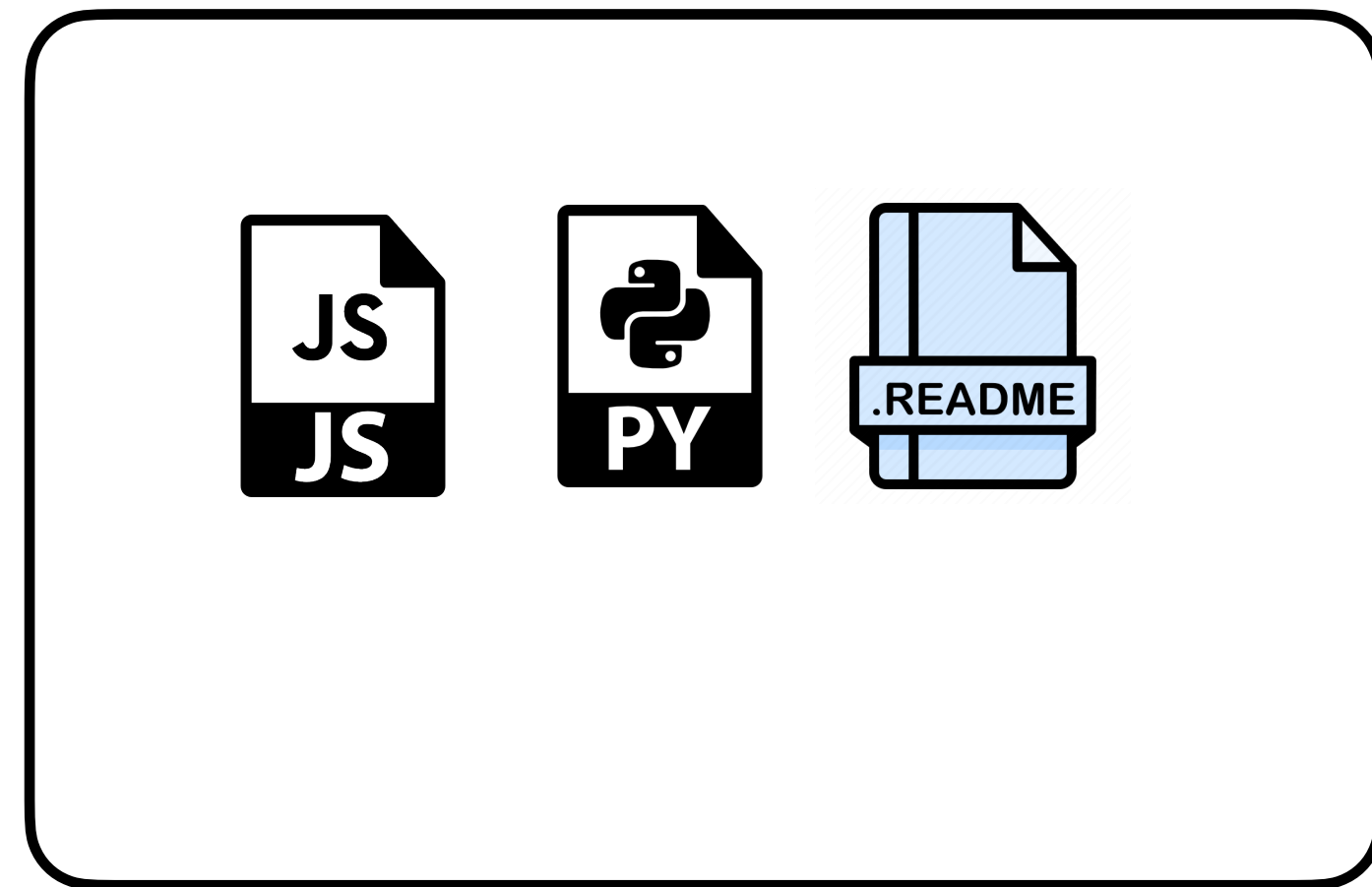
General git workflow

Working directory



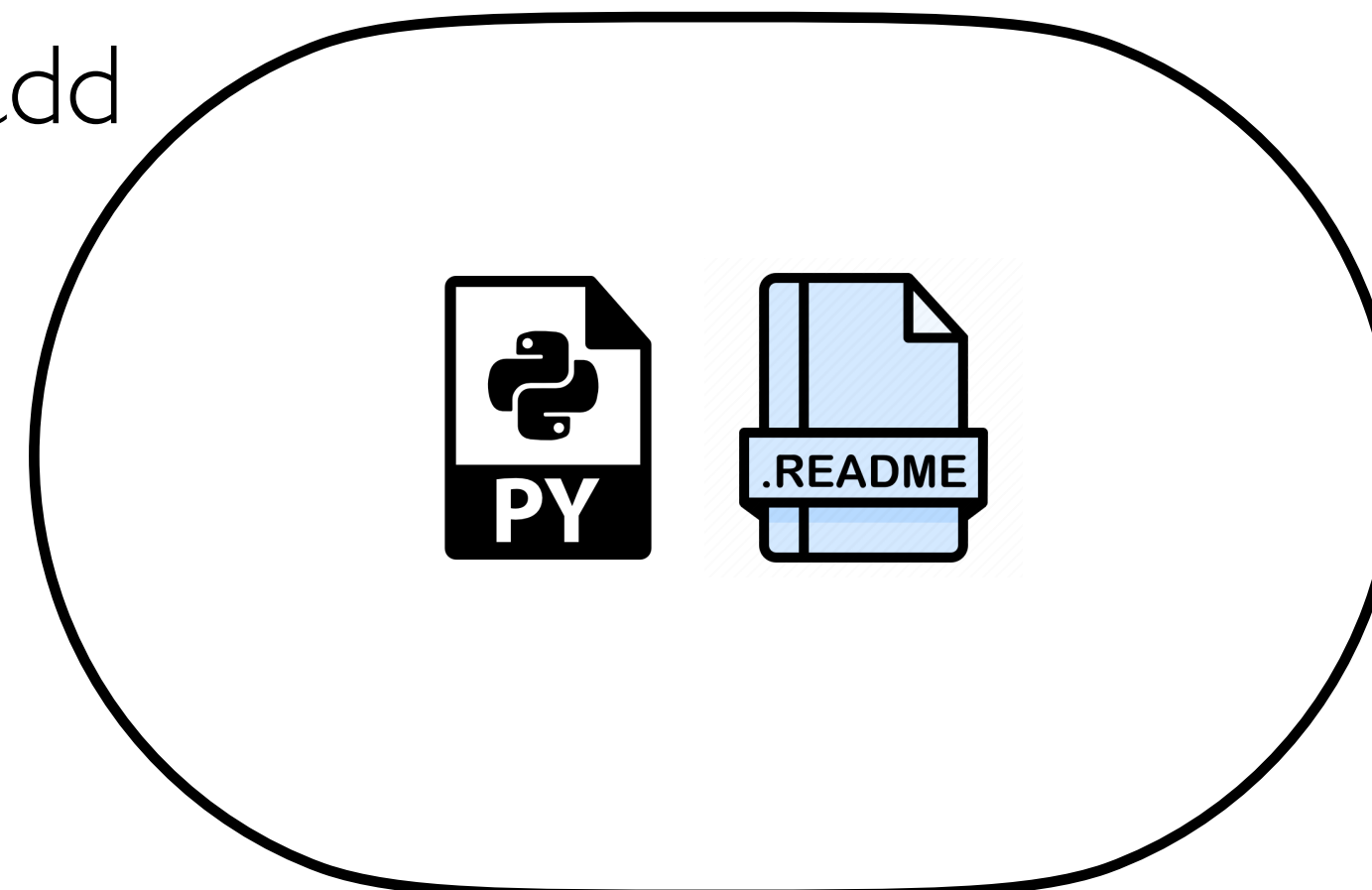
General git workflow

Working directory



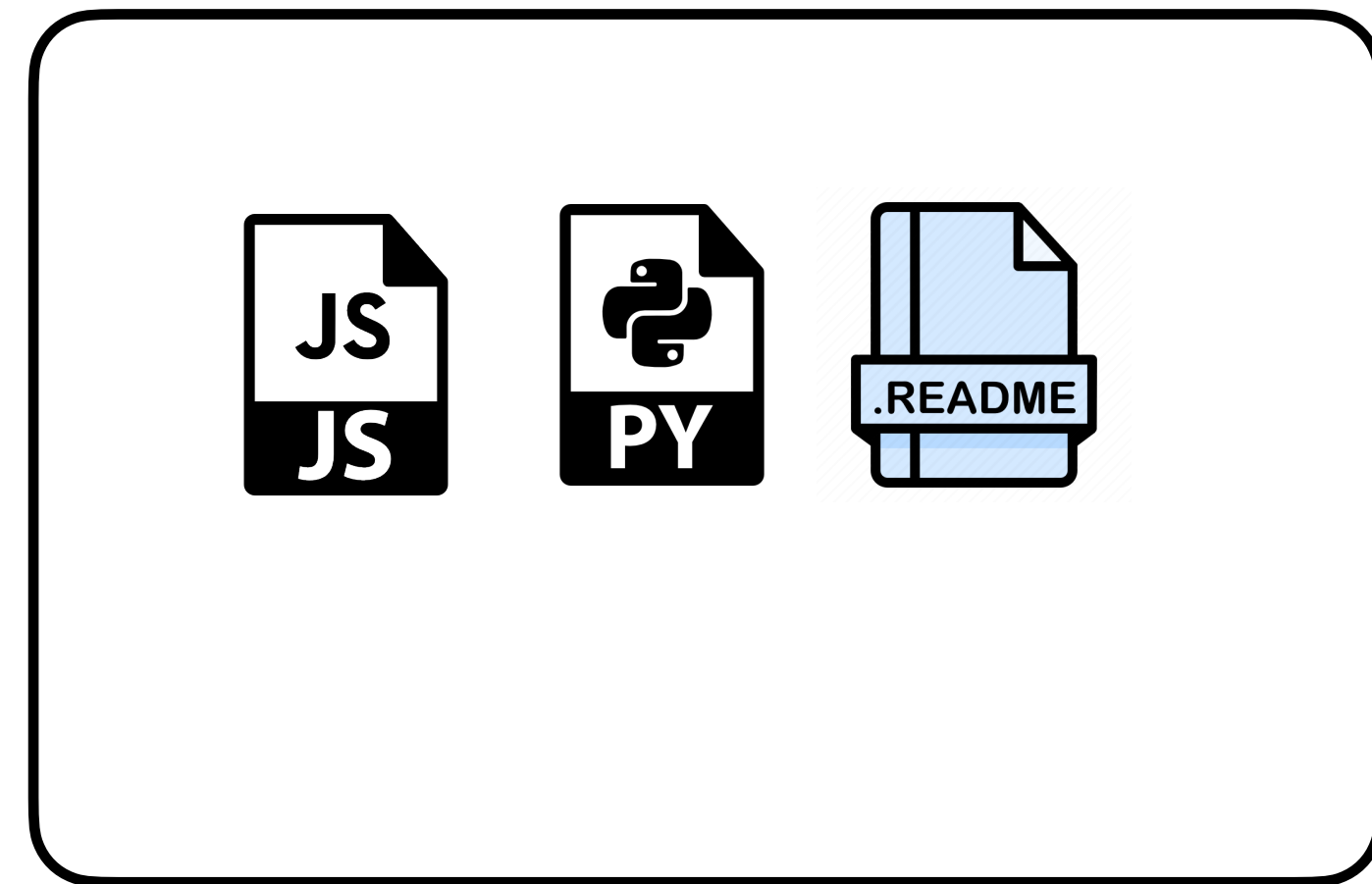
Git add

Staging or Index (old name)



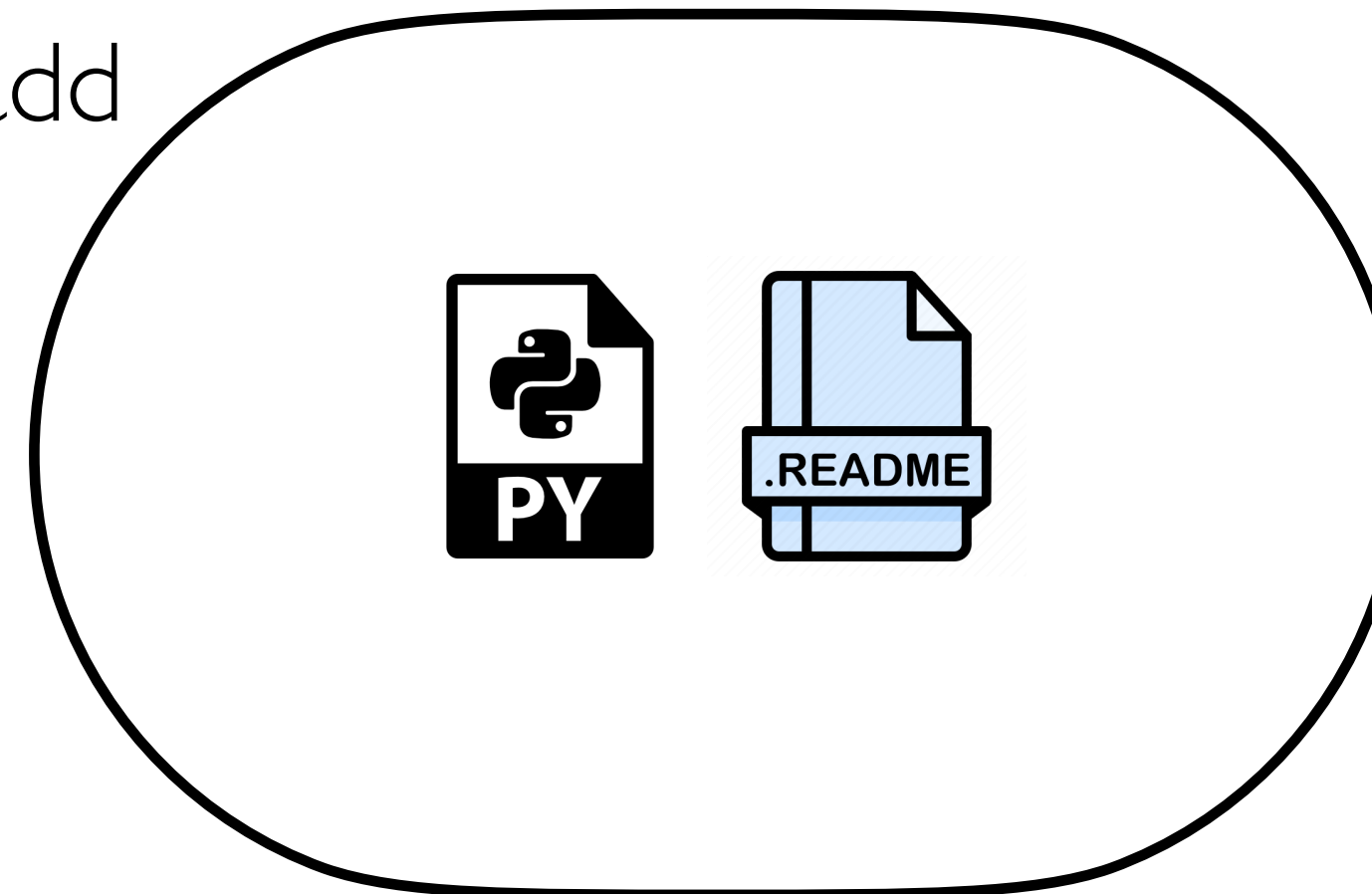
General git workflow

Working directory

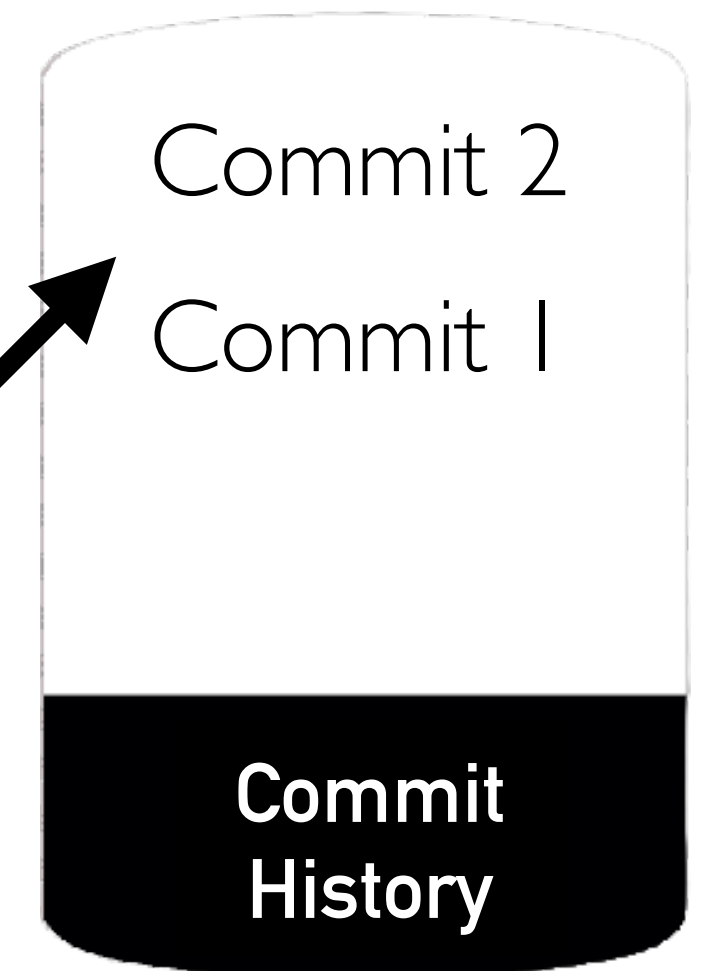


Git add

Staging or Index (old name)

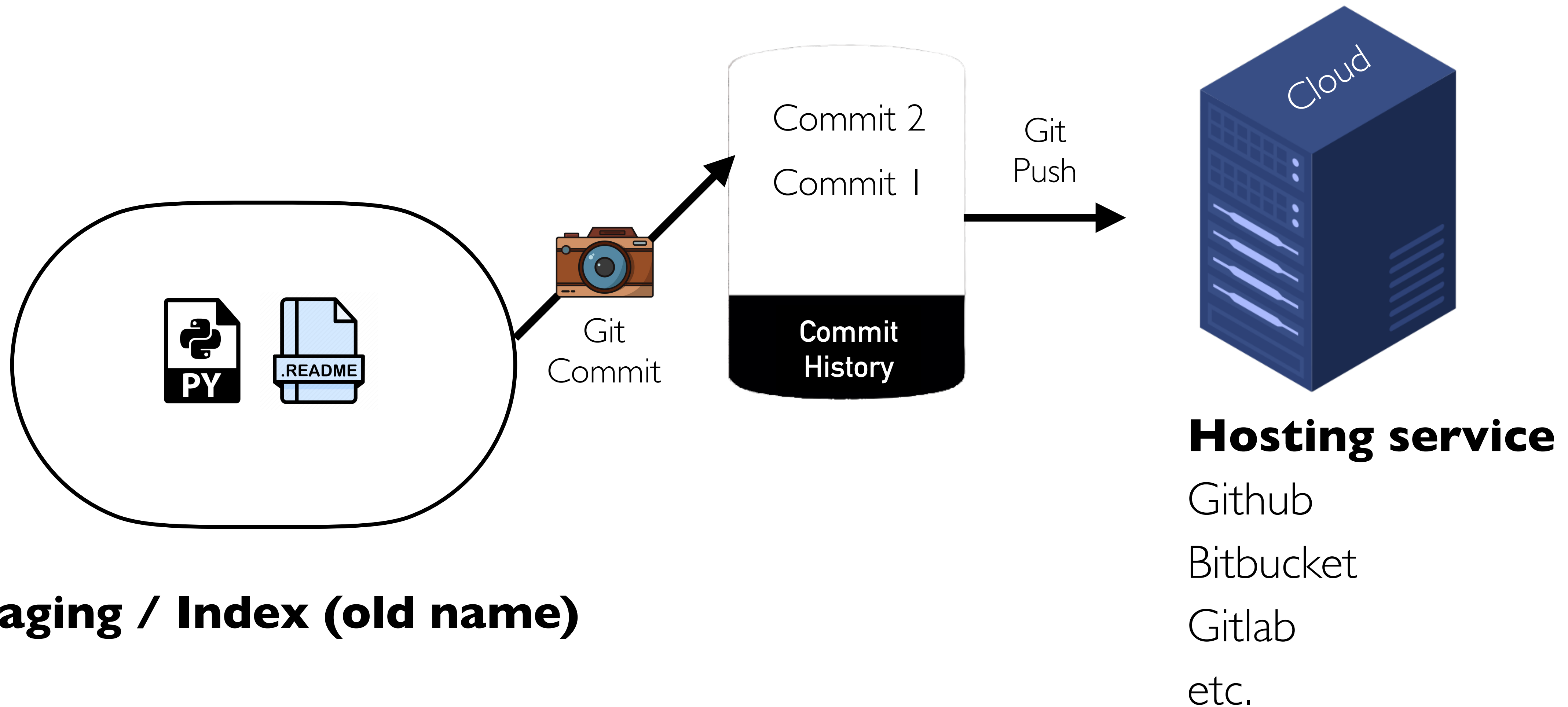


Git
Commit



Ps. Git doesn't save the same files for each commit. Files are compressed and saved uniquely

General git workflow



Git installation

- ◆ Installation instructions : <https://github.com/git-guides/install-git>
- ◆ For windows use Git BASH

```
$ git version  
git version ****
```

Git configuration

Cf. <https://git-scm.com/book/sv/v2/Customizing-Git-Git-Configuration>

```
$ git config [config level] [config name] [config value]
```

◆ [config levels] :

--local : (default) applies to the local repository, config is found in .git/config

--global : applies to the OS user repositories, config is found in ~/.gitconfig

--system : applies to the all OS users, config is found in /etc/gitconfig

◆ [config Name] :

user : email | name

core : editor

alias : git command

```
$ git config --global user.name 'amine'
```

```
$ git config --global user.email 'youremail@domain.com'
```

```
$ git config --local alias.st status -s
```

```
$ git config --global alias.lg 'log --oneline --graph --all'
```

```
$ git config --global -e #To see the current configuration
```

Bash commandes reminder

\$ cd

\$ ls -A

\$ pwd

\$ chmod

\$ grep

\$ mv

\$ cp

\$ rm -r

\$ mkdir

\$ touch

\$ cat

\$ diff

\$ echo

\$ man

\$ history

\$ clear

\$ locate

\$ open .

\$ exit

\$ kill

\$ head

\$ tail

\$ ps

Initialise git

```
$ git init .
```

```
$ mkdir "hello git" && cd "$_"
```

```
$ git init .
```

```
$ ls -a
```

```
.  ..  .git
```

Add file(s) to the staging area

```
$ git add <file>  
$ git ls-files #like ls on stage
```

- ◆ <file> can be :
 - ▶ String : File name
 - ▶ • : all file within directory (Not recommended)
 - ▶ * • [extension]

```
$ touch main.py  
$ echo "print('hello world')" > main.py  
$ cat main.py  
$ git add main.py
```

Make a commit (Take snapshot)

```
$ git commit -m "your commit message here"
```

```
$ git commit -m "Initial commit."
```

Change (last) commit message

```
$ git commit --amend -m "your new commit message"
```

You can commit and add on the same time (adds files already tracked to the staging)

```
$ git commit -am "your new commit message"
```

Git log

Log history of commits

```
$ git log  
$ git log --oneline  
$ git log --oneline --reverse #start from the first commit
```

Commit hash

Current branch

Commit message

```
[macbook-pro-de-amine:hello git amine$ git log  
commit 61ad5ba39f8dccdae64b5b1a1825bbb2138cc0b6 (HEAD -> main)  
Author: AmineFrj <ferdjaouiamine@gmail.com>  
Date: Mon Nov 28 14:12:48 2022 +0100  
  
    Rename main.py  
  
commit fb90b43d2d78c2c709da49afe8e65e7d40e6d69f  
Author: AmineFrj <ferdjaouiamine@gmail.com>  
Date: Mon Nov 28 14:06:22 2022 +0100  
  
    add readme
```

Git show & HEAD

View commit details

```
$ git show <hash>  
$ git show HEAD~N #The N-th commit before head  
$ git show HEAD~N:<filename>  
$ git ls-tree HEAD~N #like ls -a on certain commit
```

Head is a pointer to the a branch (branche's latest commit)*

*The HEAD could also be pointing to another commit and in that case it is called *Detached HEAD*

Check changes

```
$ git status
```

```
$ git status -s #cf. https://git-scm.com/docs/git-status
```

```
$ git status
```

```
$ echo "print('Here is a new line')" >> main.py
```

```
$ git status
```

```
$ git add main.py
```

```
$ git status
```

```
$ git commit -m "Add new print"
```

Delete file(s)

Remove file(s) from working dir and in staging

```
$ git rm <file>
```

```
$ echo "test" >> README.txt
```

```
$ git add README.txt
```

```
$ git commit -m "Add a README."
```

```
$ git rm README.txt
```

```
$ git status
```

```
$ git commit -m "Remove README"
```

Delete file(s) from staging

Remove file(s) from staging

```
$ git rm --cached <file>
```

```
$ git ls-files
```

```
$ touch main.py
```

```
$ echo "print('hello world')" > main.py
```

```
$ cat main.py
```

```
$ git add main.py
```

Rename file(s)

```
$ git mv <old_name> <new_name>
```

```
$ git mv main.py model.py
```

```
$ git status
```

```
$ git commit -m "Rename main.py"
```

```
$ git status
```

Ignore files

Cf. <https://github.com/github/gitignore>

```
$ vi .gitignore
```

```
$ echo requirements.txt > .gitignore
```

Ps. Files created before .gitignore will not be ignored, you have to remove them from the staging

```
$ git rm --cached <file>
```

Then commit the changes

View Staged differences

\$ git diff #Difference between working dir and staging

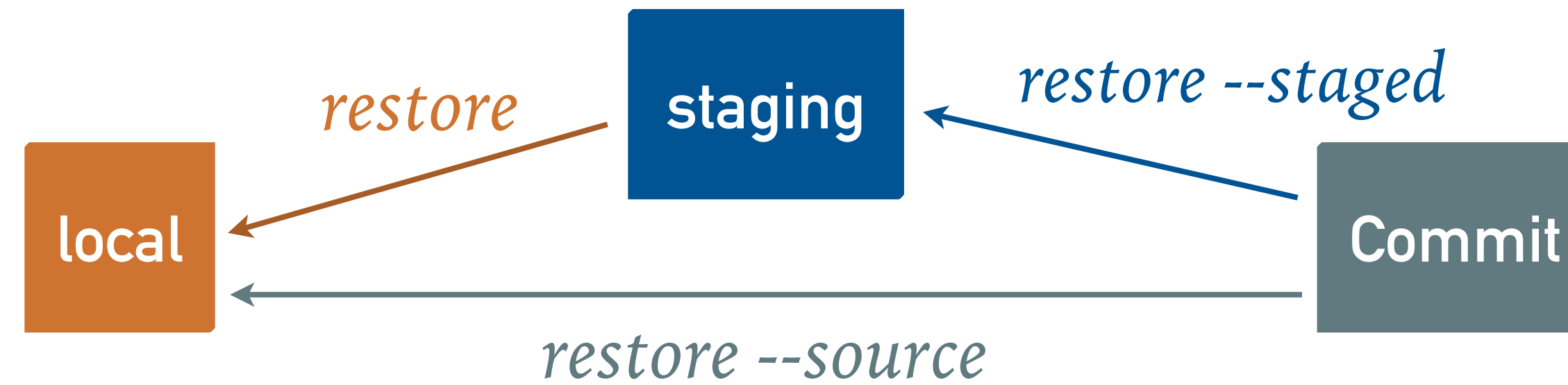
\$ git diff --staged #Difference between staging and last commit

The image shows a terminal window with the command `git diff --staged` executed. The output shows the difference between the last commit and the current staging area. Annotations with red arrows point to specific parts of the output:

- Last commit**: Points to the commit hash `7f58f33` in the line `index 7f58f33..411db58 100644`.
- Current staging**: Points to the commit hash `411db58` in the same line.
- Staging unique id**: Points to the file path `b/main2.py` in the line `diff --git a/main2.py b/main2.py`.
- The 3 lines extracted from last commit**: Points to the range `-1,3` in the line `@@ -1,3 +1,4 @@`.
- New line**: Points to the new line `+#thisisacomment` in the output.

```
[macbook-pro-de-amine:hello git amine$ git diff --staged
diff --git a/main2.py b/main2.py
index 7f58f33..411db58 100644
--- a/main2.py
+++ b/main2.py
@@ -1,3 +1,4 @@
 print('hello world')
 print("Here is a new line \!")
 print(Here is a new line)
+#thisisacomment
macbook-pro-de-amine:hello git amine$
```

Some commands to cancel changes



Cancel local changes : go back to the last staged files

```
$ git restore <file>
```

Cancel staged files : go back to the previous commit

```
$ git restore --staged <file>
```

Restore local files from commit

```
$ git restore --source=HEAD <file>
```

```
$ git restore --source=HEAD~[N] <file>
```

Best practices

- ◆ Commit messages should be meaningful, not too short nor too long
- ◆ Use present tense in commit message (for exp. "Fix the bug")
- ◆ Avoid using force command as much as possible (the -f flag)
- ◆ Use the terminal instead of service providers web pages

1. Créer un nouveau dossier avec deux fichiers (main.py et requirements.txt) et initialiser un git
2. Regarder le status de votre fichier et du staging
3. Ajouter main.py au staging et regarder à nouveau l'état du staging
4. Créer un commit
5. Regarder la liste des commits
6. Regarder le contenu du dernier commit
7. Mettre à jour le main.py et regarder le status de votre dossier
8. Ignorer le requirements
9. Regarder le status de votre dossier
10. Créer un fichier preprocessing.py et l'ajouter dans le staging
11. Annuler le stage